Careers in Data Science

Introduction

Data science is an interdisciplinary field that uses statistics, mathematics, programming, machine learning and domain knowledge to extract useful insights and predictions from data. Businesses, governments, hospitals, banks and sports teams collect huge amounts of data, and data science turns it into decisions: which customers might leave, which transactions are fraud, which route is fastest or which patients are at risk. The phrase "data scientist" was popularised around 2008 by DJ Patil and Jeff Hammerbacher, and in 2012 the Harvard Business Review called it "the sexiest job of the 21st century". Data science is among the most in-demand careers in India and the world, and offers careers in IT services, e-commerce, banking, healthcare, manufacturing, telecom and research. Chennai, Bengaluru, Hyderabad and Pune have many data science roles.

What Does a Data Scientist Do?

A data scientist collects, cleans and analyses large and complex data, builds statistical and machine learning models to find patterns and make predictions, and communicates results to decision-makers. The typical work cycle has these steps: understanding the business problem; collecting data from databases, APIs, sensors, logs and files; cleaning the data by handling missing values, errors and duplicates, which often takes the largest part of the time; exploring the data with statistics and visualisation (exploratory data analysis); building and training models, such as regression, classification, clustering, time-series forecasting, deep learning and natural language processing; evaluating them with metrics; deploying them into applications; and monitoring performance. They also design experiments such as A/B tests and tell stories with data through dashboards and presentations. The CRISP-DM framework is a common process model.

Name Three Key Skills

Three key skills for a data scientist are programming (especially Python, plus SQL), statistics and mathematics, and machine learning. Python is the leading language, with libraries such as NumPy, Pandas, Scikit-learn, Matplotlib, TensorFlow and PyTorch; R is also used, and SQL is essential to retrieve data from databases. Statistics and probability (distributions, hypothesis testing, regression), linear algebra and calculus provide the foundation for understanding models. Machine learning covers supervised and unsupervised algorithms, model evaluation and increasingly deep learning and generative AI. Other important skills include data visualisation and storytelling with tools like Tableau, Power BI and Matplotlib; data wrangling; big data tools like Spark; cloud platforms such as AWS, Azure and Google Cloud; version control with Git; domain knowledge of the industry; and soft skills such as communication, curiosity, critical thinking and teamwork. The ability to explain results to non-technical people is as important as technical ability.

What Is the Difference Between a Data Analyst and a Data Scientist?

A data analyst examines existing data to answer specific business questions, describing what has happened and why, using tools such as Excel, SQL and BI tools such as Power BI and Tableau, along with basic statistics, and produces reports and dashboards. The focus is on descriptive and diagnostic analytics, and the role usually requires less programming and advanced mathematics. A data scientist goes further: they work with larger, messier and sometimes unstructured data (text, images), use advanced statistics, programming and machine learning to build predictive and prescriptive models that forecast what will happen and recommend actions, design experiments, and often work on open-ended problems. In short, the analyst explains the past and present, while the scientist predicts the future and builds models, although the roles overlap, and many analysts grow into scientists. Related roles include the data engineer, who builds the pipelines and infrastructure that move and store data; the machine learning engineer, who deploys models into production; the business intelligence developer; the AI research scientist; the MLOps engineer; and the data architect.

How to Become a Data Scientist

Most people begin with a bachelor's degree in computer science, engineering, mathematics, statistics or a related field, though people from other backgrounds also move in. A master's degree is useful for some roles, but not always necessary. In India, IIT Madras offers an online BS degree in Data Science and Applications, launched in 2020, which is open to students from different backgrounds. Free and low-cost courses are available on NPTEL, Coursera, edX and Kaggle Learn. A strong portfolio is vital: do real projects, take part in Kaggle competitions, publish code on GitHub, write blog posts and do internships. Steps include learning Python and SQL, statistics, machine learning and visualisation, building three or four projects on real datasets, learning cloud and deployment basics, and preparing for interviews with practice in SQL, coding, statistics and case studies. Freshers often start as data analysts or junior data scientists.

Industries and Opportunities

Data scientists work in e-commerce (recommendation engines at Flipkart and Amazon), banking (credit scoring and fraud detection), healthcare (diagnosis from scans), agriculture (yield prediction), manufacturing (predictive maintenance, quality inspection), telecom, media (content recommendations), sports analytics and government. Chennai's IT, automotive, healthcare and startup sectors, with companies such as Zoho, Freshworks, HCL, TCS, Cognizant and many global capability centres, offer roles in data and AI. Generative AI and large language models have created new roles such as prompt engineer, AI engineer and LLM application developer.

Challenges and Future

The field is competitive, and tools change quickly, so continuous learning is essential. Automated machine learning (AutoML) and AI assistants are automating routine tasks, so the focus is moving to problem formulation, domain understanding, ethics and communication. Data privacy laws such as India's Digital Personal Data Protection Act, 2023 and responsible AI practices are important.

Quick Facts

- A data scientist collects, cleans and analyses data and builds models to extract insights and make predictions.
- Key skills: programming (Python and SQL), statistics and mathematics, and machine learning; also data visualisation and communication.
- A data analyst explains what happened using existing data; a data scientist builds predictive models and works with more complex data.
- IIT Madras launched an online BS degree in Data Science and Applications in 2020.

- Harvard Business Review called data scientist the sexiest job of the 21st century in 2012.